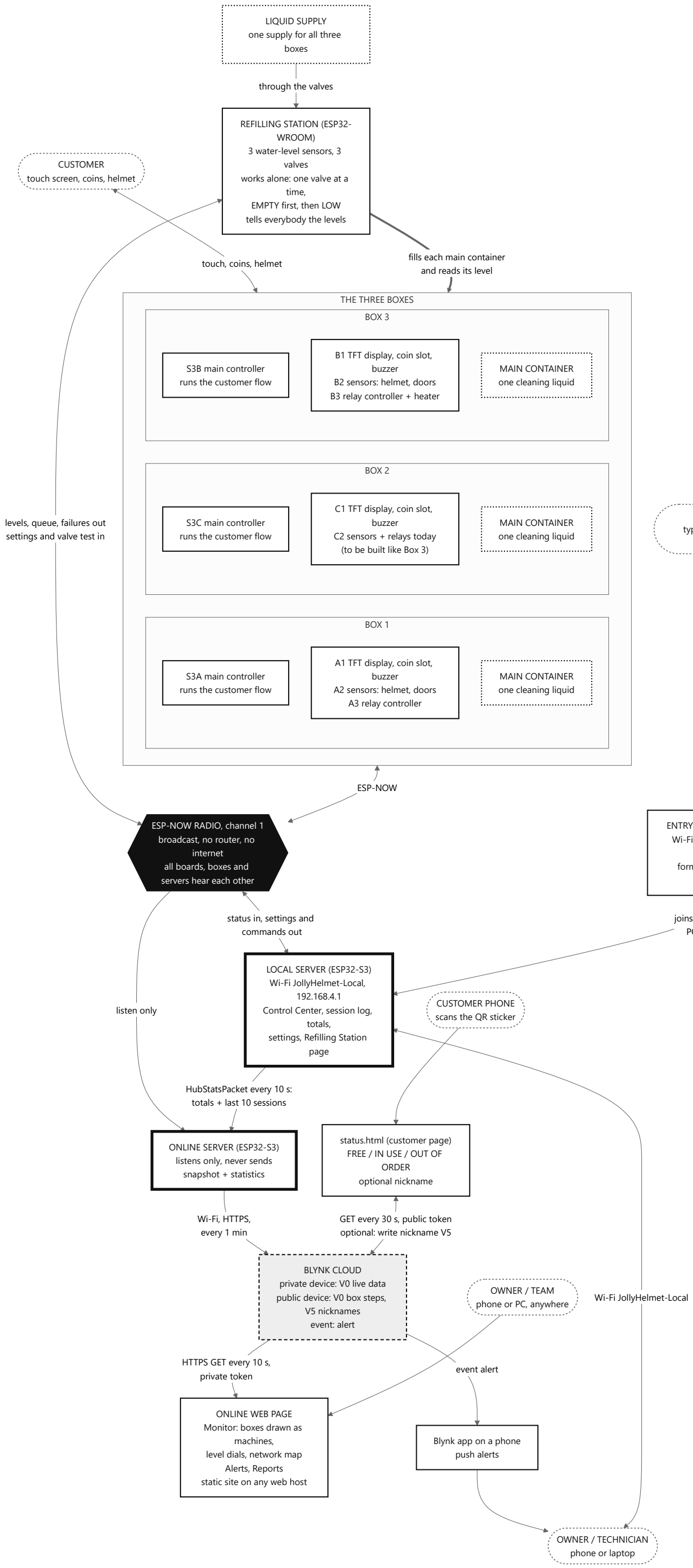
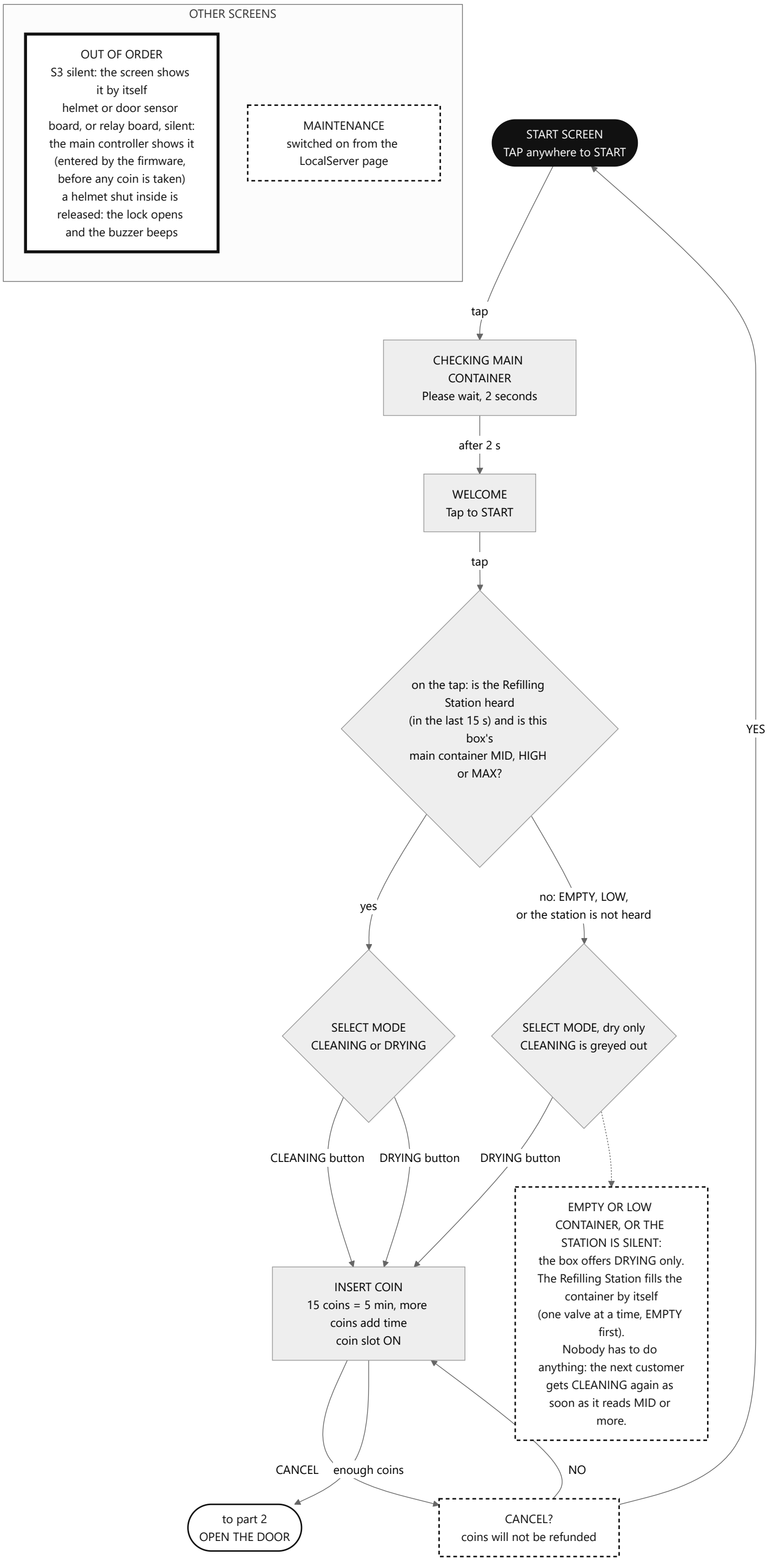
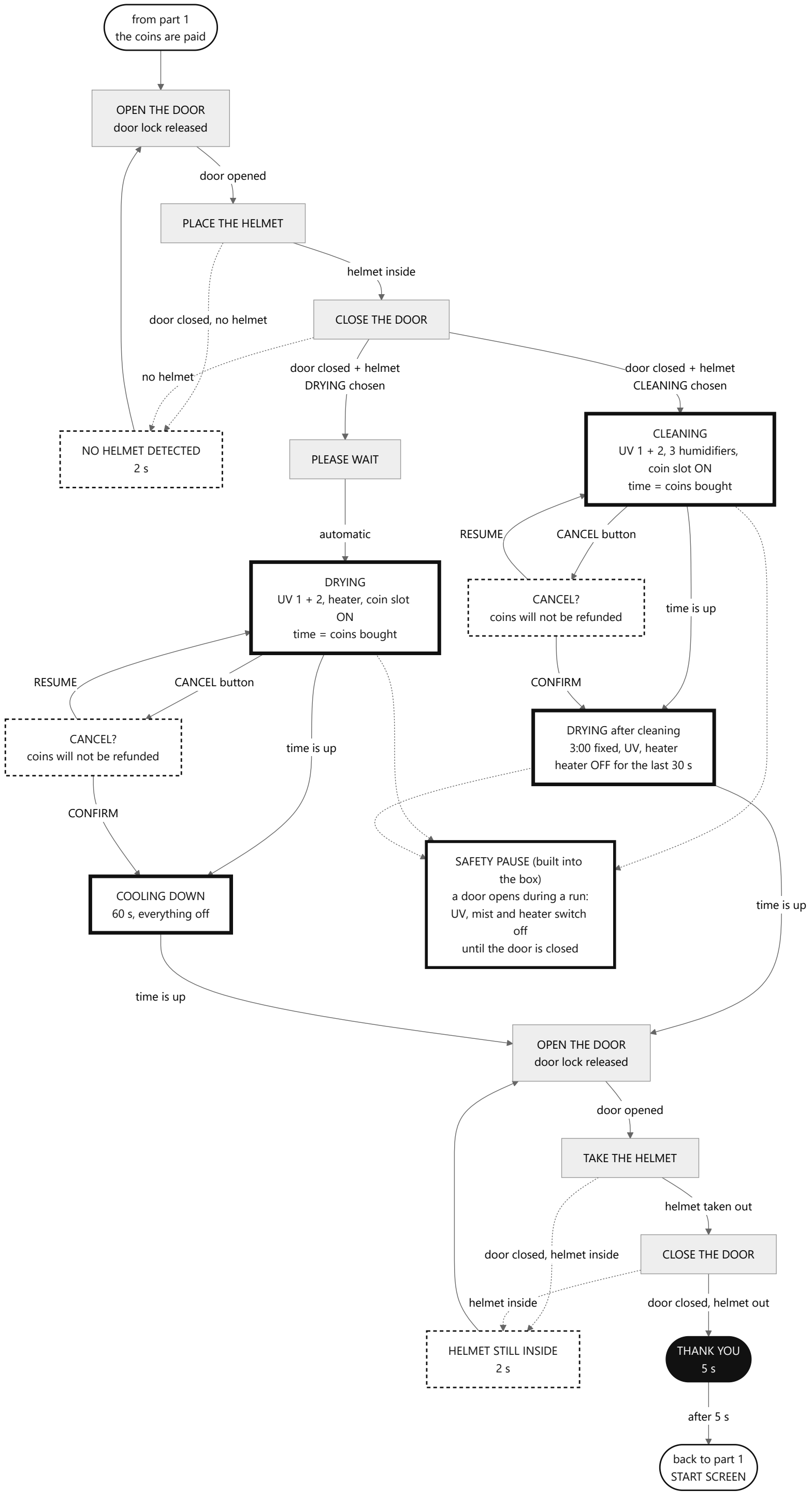
01 - The whole system (three boxes, the Refilling Station, the servers)

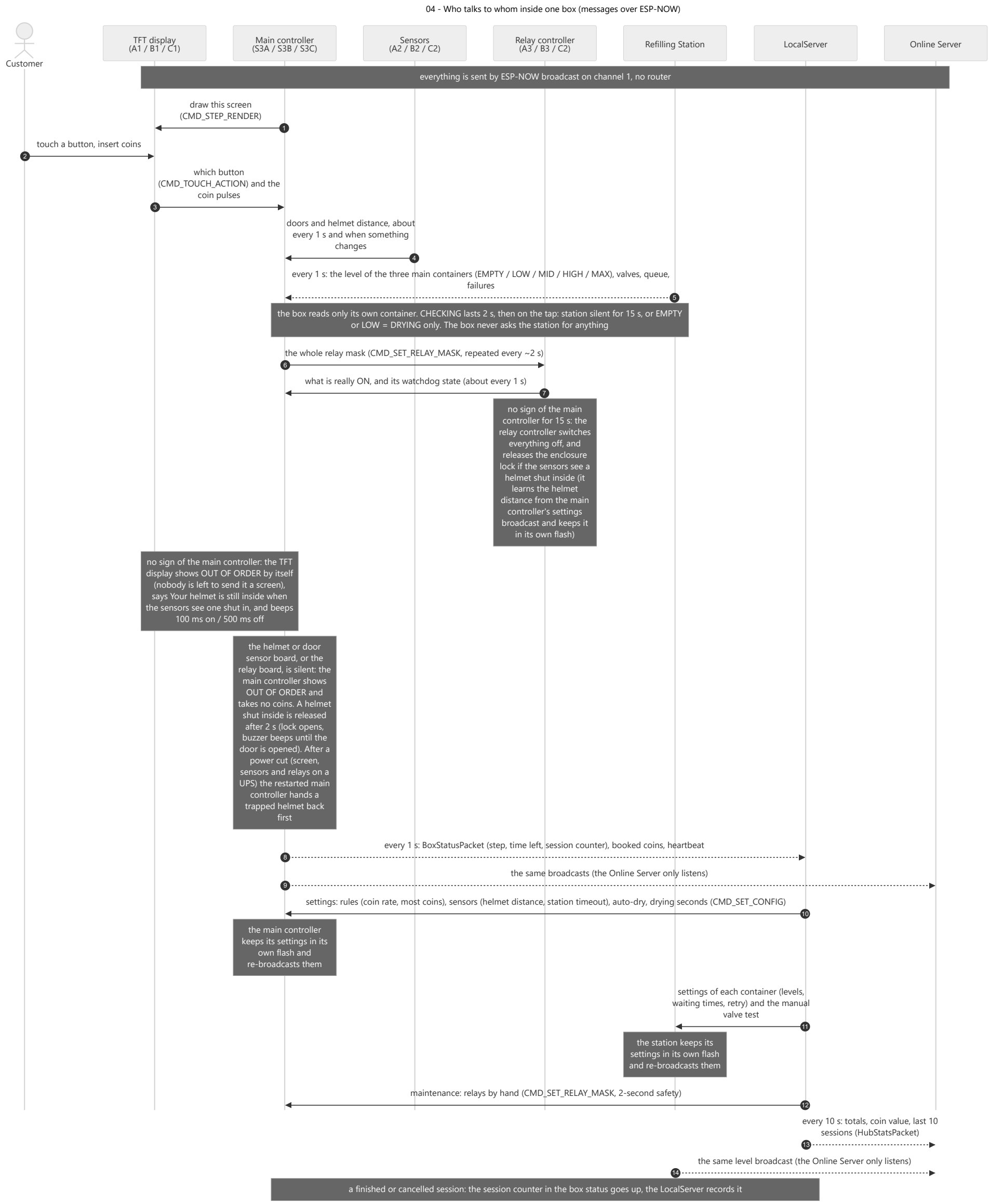


02 - Customer flow of a box, part 1 - start, checking the container, mode, coins (same for Box 1, 2 and 3)

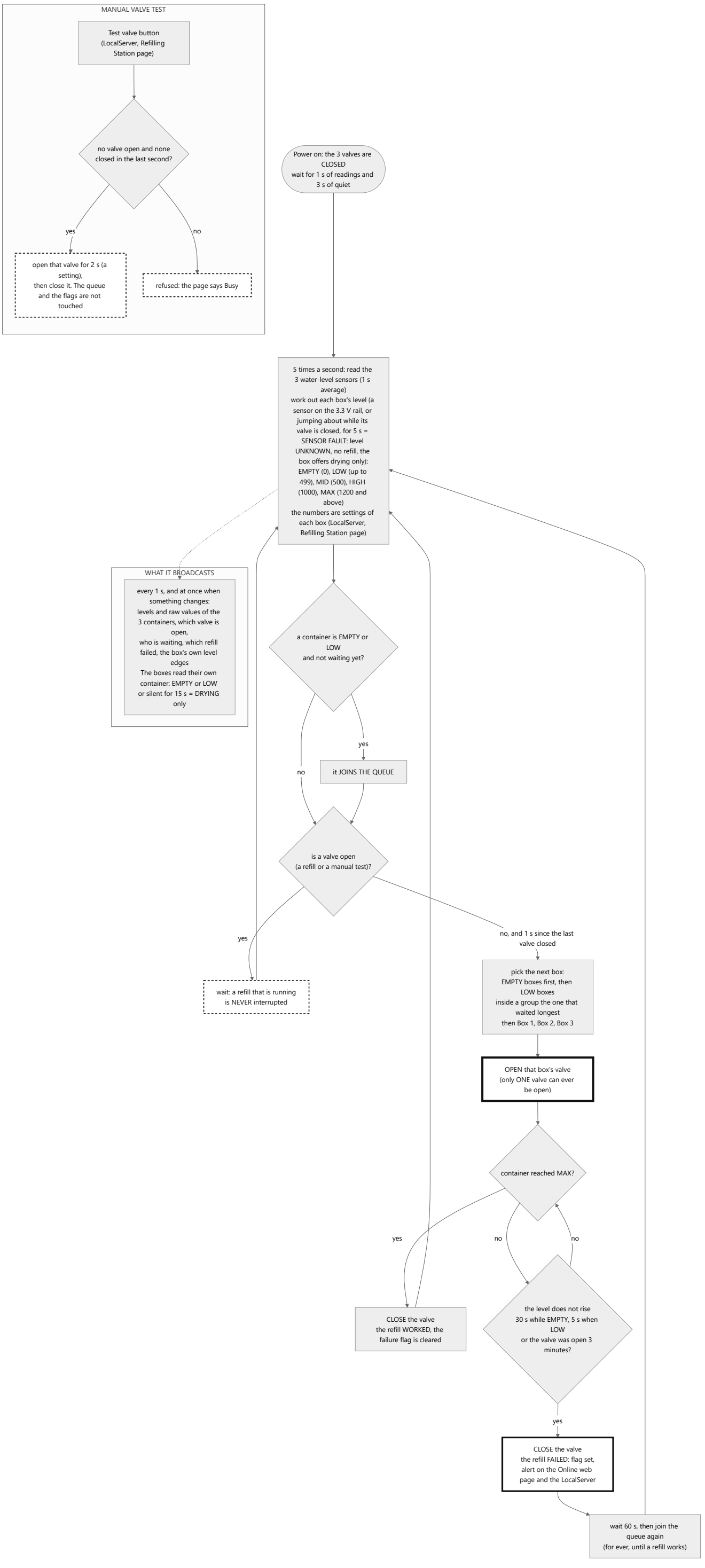


03 - Customer flow of a box, part 2 - door, run, drying, finish (same for Box 1, 2 and 3)

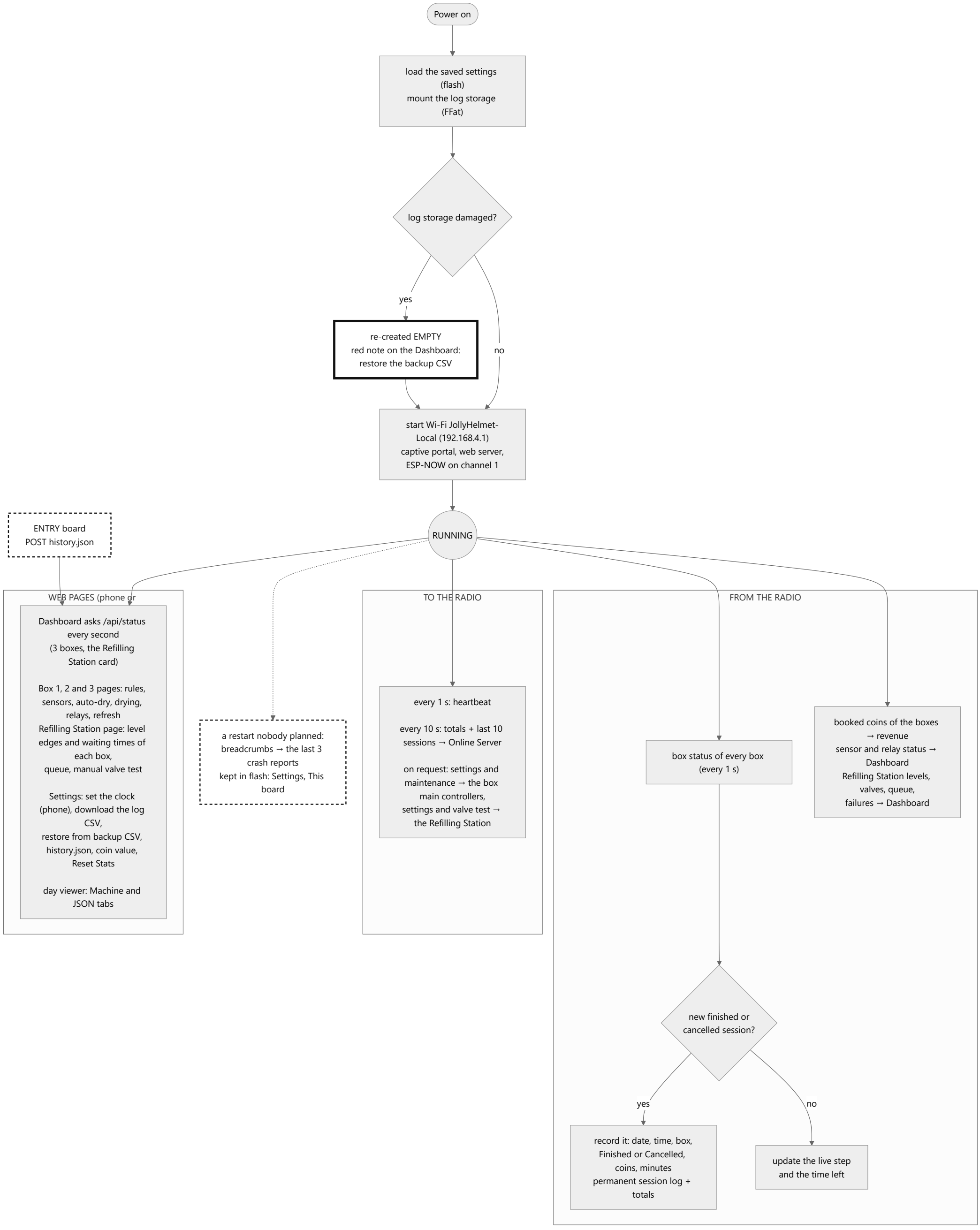



04 - Who talks to whom inside one box (messages over ESP-NOW)
Customer
TFT display (A1 / B1 / C1)
Main controller (S3A / S3B / S3C)
Sensors (A2 / B2 / C2)
Relay controller (A3 / B3 / C2)
Refilling Station
LocalServer
Online Server
everything is sent by ESP-NOW broadcast on channel 1, no router
draw this screen (CMD_STEP_RENDER)
1
touch a button, insert coins
2
which button (CMD_TOUCH_ACTION) and the coin pulses
3
doors and helmet distance, about every 1 s and when something changes
4
every 1 s: the level of the three main containers (EMPTY / LOW / MID / HIGH / MAX), valves, queue, failures
5
the box reads only its own container. CHECKING lasts 2 s, then on the tap: station silent for 15 s, or EMPTY or LOW = DRYING only. The box never asks the station for anything
the whole relay mask (CMD_SET_RELAY_MASK, repeated every ~2 s)
6
what is really ON, and its watchdog state (about every 1 s)
7
no sign of the main controller for 15 s: the relay controller switches everything off, and releases the enclosure lock if the sensors see a helmet shut inside (it learns the helmet distance from the main controller's settings broadcast and keeps it in its own flash)
no sign of the main controller: the TFT display shows OUT OF ORDER by itself (nobody is left to send it a screen), says Your helmet is still inside when the sensors see one shut in, and beeps 100 ms on / 500 ms off
the helmet or door sensor board, or the relay board, is silent: the main controller shows OUT OF ORDER and takes no coins. A helmet shut inside is released after 2 s (lock opens, buzzer beeps until the door is opened). After a power cut (screen, sensors and relays on a UPS) the restarted main controller hands a trapped helmet back first
every 1 s: BoxStatusPacket (step, time left, session counter), booked coins, heartbeat
8
the same broadcasts (the Online Server only listens)
9
settings: rules (coin rate, most coins), sensors (helmet distance, station timeout), auto-dry, drying seconds (CMD_SET_CONFIG)
10
the main controller keeps its settings in its own flash and re-broadcasts them
settings of each container (levels, waiting times, retry) and the manual valve test
11
the station keeps its settings in its own flash and re-broadcasts them
maintenance: relays by hand (CMD_SET_RELAY_MASK, 2-second safety)
12
every 10 s: totals, coin value, last 10 sessions (HubStatsPacket)
13
the same level broadcast (the Online Server only listens)
14
a finished or cancelled session: the session counter in the box status goes up, the LocalServer records it

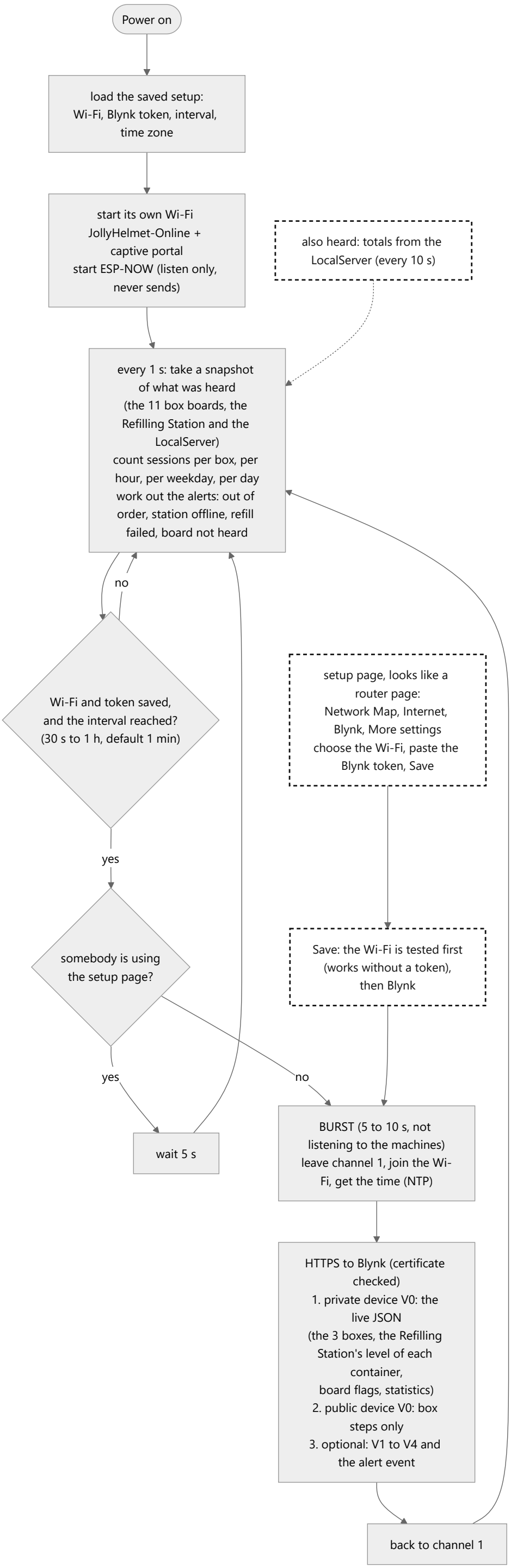
05 - The Refilling Station (one valve at a time, it works alone)



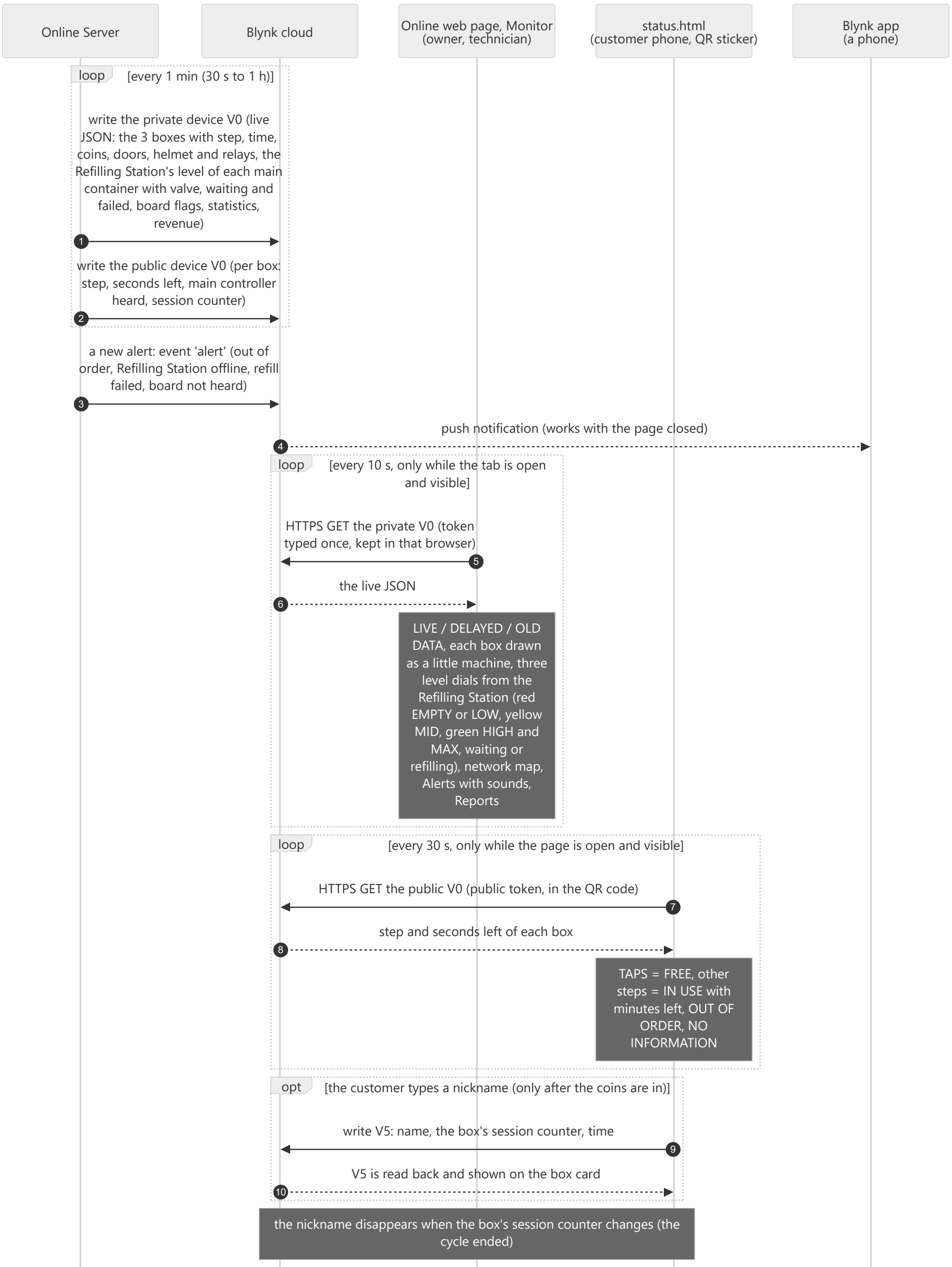
06 - LocalServer (Control Center) inside



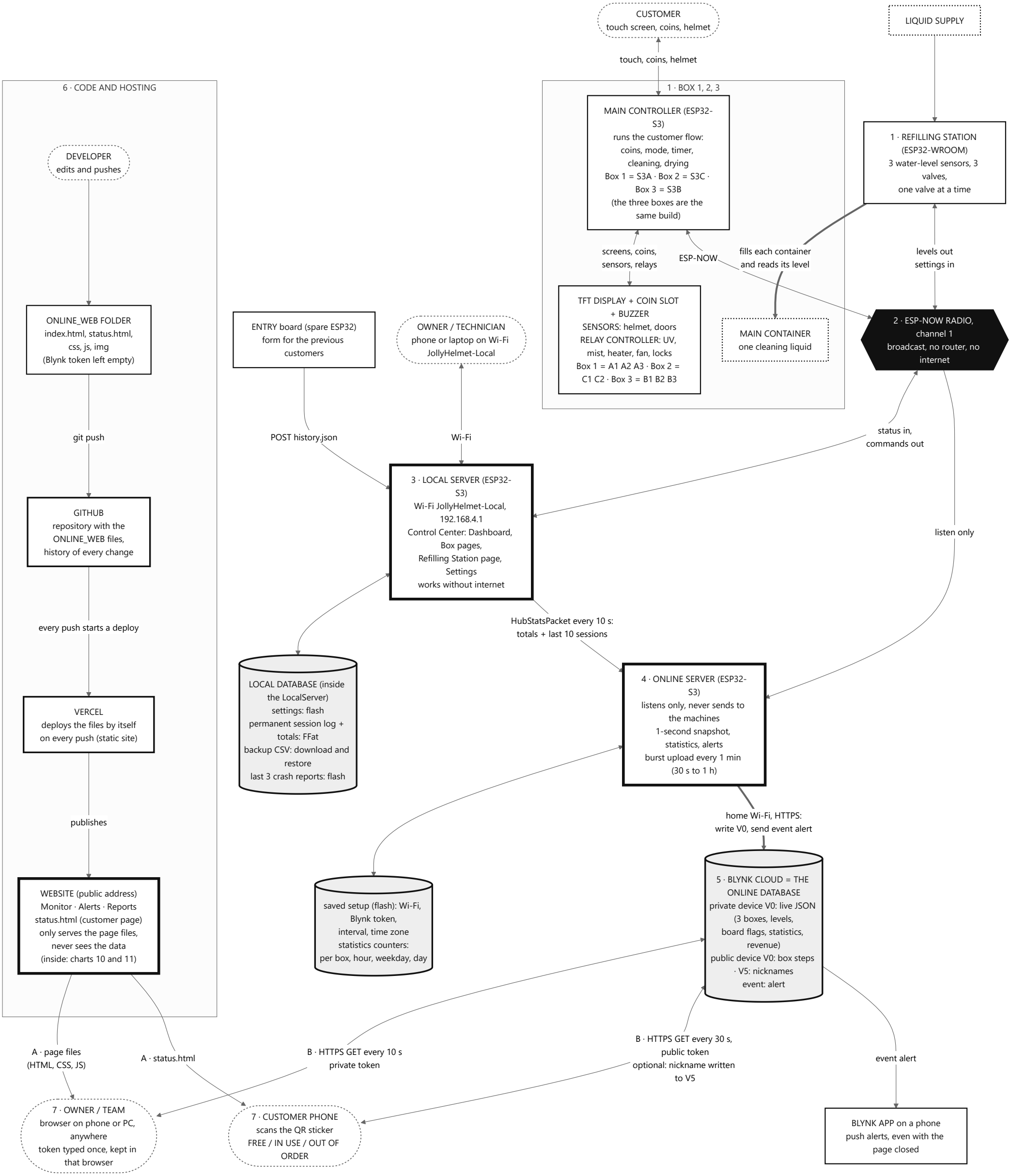
07 - Online Server inside



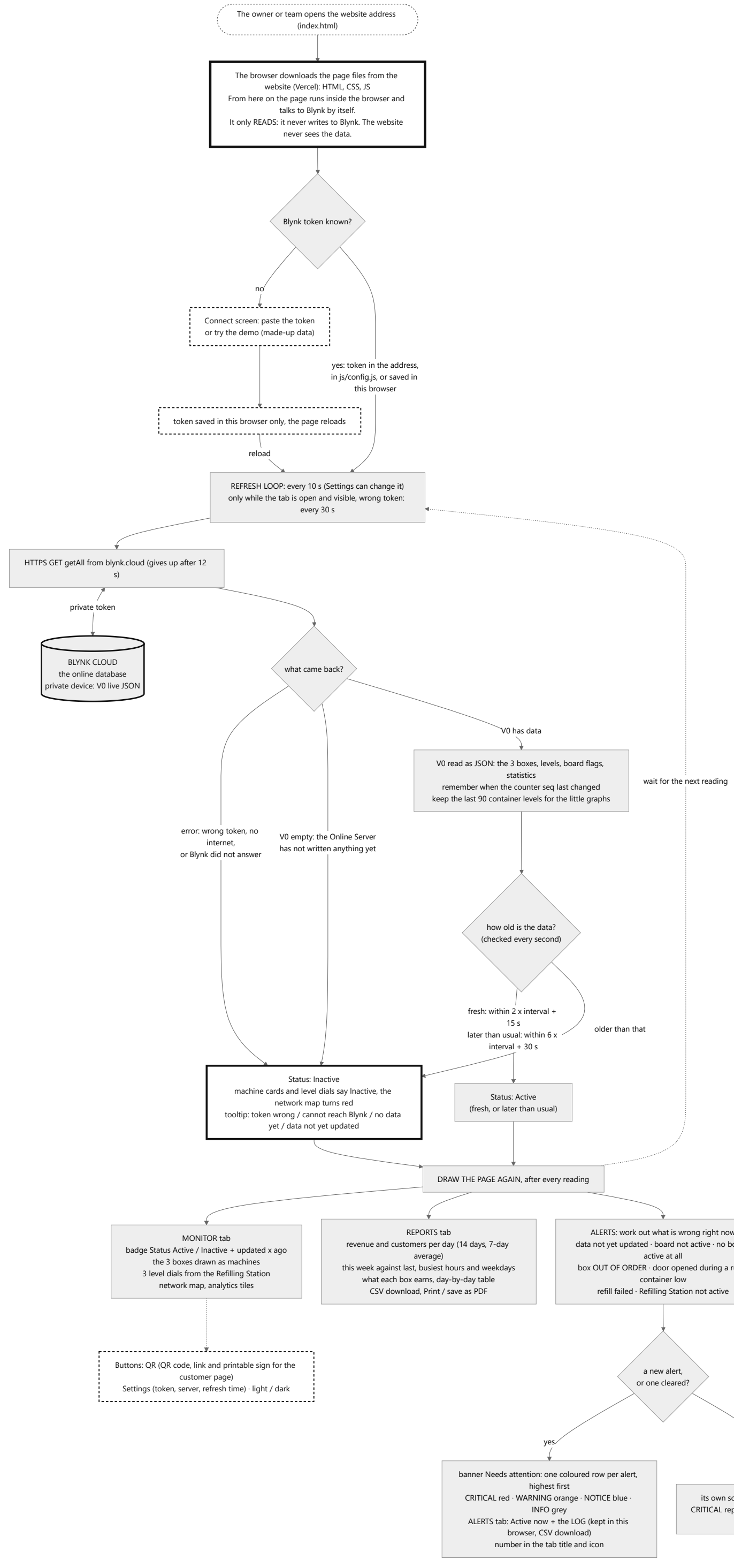
08 - Blynk and the web pages



09 - The complete Jolly Helmet system, end to end (hardware, LocalServer, Online Server, database, GitHub, Vercel, website)



10 - The website inside, part 1 - the Monitor (index.html, for the owner and the team)



11 - The website inside, part 2 - the customer page (status.html, from the QR sticker)

